Review: FS-PFAN Laparoscopic Desmoking Privacy (DA-3)

Overall verdict

A careful, honest plan, but still only a plan. It needs at least one real experiment before it is a paper.

What is good: the writing is precise about what is and is not claimed. It separates video identity from patient identity, uses video-disjoint splits, includes a compute-matched control, and states hypotheses that can fail. The minimax sign in Section III-B is also correct, which many students get wrong.

The problem: there are no results at all. Table II is all "TBD", and the abstract says "No FS-PFAN experiments have yet been run." Conferences almost never accept a deep-learning paper that is only a protocol. For DA-3 it is fine as a progress report, but not yet a paper.

Major comments

1. Check the data first; the study may not be possible as written

The study needs the source video ID of every frame. The paper itself says the PFAN 660-pair data may not have this mapping ("must be confirmed"). Check this now, before anything else.

- If the PFAN repository gives the Cholec80 video ID for each frame, go ahead.
- If not, don't wait. Build your own pairs: sample Cholec80 frames yourselves, so you know each video ID, and add smoke with PFAN's Blender method. This also gives you more data.

2. Too few images per video

660 pairs over up to 80 videos is about 8 frames per video. Telling 80 videos apart from about 8 images each, and fewer after splitting, gives noisy, unreliable numbers. Write down how many videos and frames per video you will have in each split. Increase sampling if needed (for example, every 5 s instead of 20 s).

3. The attack setup contradicts the split

With video-disjoint splits the test videos are new, but a "which video is this?" classifier can only name videos it has seen. Section IV-C notices this but lists two options without choosing one.

My suggestion: change the attack to **linkage**. Show the attacker two frames and ask, "Are these from the same video?" This works on unseen videos, fits the video-disjoint split, and matches the real risk (linking a shared frame to a recording) better than closed-set classification.

4. Explain why this is a real privacy risk

A reviewer will ask: who is harmed if someone learns which Cholec80 recording a desmoked inside-the-body frame came from? Add one short, concrete scenario, for example: frames are shared for research, and linking them back to a hospital recording reveals the patient's surgery details.

Also discuss the built-in conflict. Much of the video identity is in the anatomy itself (this patient's organs and tissue), and a desmoking model must keep the anatomy. So removing identity may mean damaging the image. Say so, and explain why you still expect a useful trade-off; otherwise H3 looks likely to fail from the start.

5. Remove notes-to-self and AI-style instructions

Several sentences read like instructions to whoever writes the paper, not text a reader should see:

Find	Replace with
"Their exact definitions and weights must be copied from the baseline implementation and recorded; this paper does not assume unverified values."	State the actual loss weights from the PFAN code
"After the experiment, populate it with mean and … replace this paragraph with the observed outcomes. Do not describe an empty table as evidence of effectiveness."	Delete. Put real numbers in, or remove Table II
"A targeted prior-art review is still required before making a first-ever novelty claim."	Do the review, then state the novelty
"Access to the paired PFAN data and its source-frame mapping must be confirmed before analysis."	Confirm it, and report the result

6. The literature review is far too thin

There are only 8 references. A conference paper needs about 20 to 25. Add:

- other surgical desmoking methods besides PFAN (GAN-based and transformer-based desmoking on Cholec80 and similar data)
- privacy and de-identification for surgical and endoscopic video
- frequency-domain privacy methods: work that hides identity in selected frequency bands, mostly for faces. This is your closest idea and it is currently missing.

7. Minimum experiment for the next review

Run H1 only. Take pretrained PFAN, desmoke the test frames, and train a simple linkage attacker (for example, a ResNet-18 Siamese network) on smoky, clean and restored frames. Report one table of attack accuracy against chance, with confidence intervals. That turns this from a plan into a paper. If H1 fails (no identity can be recovered), that is also a valid result.

Minor comments

1. **Copyright.** Figs. 2 and 3 are copied from the PFAN paper and repository. Springer figures need permission. Redraw them in your own style and cite [1], or check the repository's license.

2. **Author block.** "1st, 2nd, 3rd, 4th" are IEEE template placeholders; remove them. The supervisor's email is missing from the 4th column. Gladwin's email has no year (gladwindaniel.p@…) while the others do; check it is correct.
3. **Abstract.** The first sentence ("Metadata sanitization does not test…") is a confusing start. Open with the problem: smoke removal, plus the risk of linking shared frames back to their recordings.
4. **Table I.** Row B6 mixes two unrelated things (compute-matched adversary and DP-SGD). Split them, or drop DP-SGD, which the paper calls optional anyway.
5. **Table captions.** In IEEE format, table captions go above the table, not below.
6. **Repetition.** Sections IV-C and V repeat the same rules on splits and intervals. Say each rule once.

Before the next review

- ☐ Confirm whether the PFAN data has video IDs; if not, build your own pairs from Cholec80
- ☐ Count videos and frames per split; increase sampling if needed
- ☐ Switch to a same-video linkage attack
- ☐ Add a one-paragraph threat scenario and discuss the anatomy-identity conflict
- ☐ Remove every notes-to-self sentence
- ☐ Expand references to about 20, including frequency-domain privacy work
- ☐ Run the H1 baseline experiment and report one real table
- ☐ Redraw Figs. 2 and 3; fix the author block

The thinking in this paper is good. It now needs data behind it.